

Problem: 9

Solve the given assignment problem
and max the sales 5x5 matrix.

68	78	50	101	82
71	84	61	73	59
87	92	111	70	81
45	64	87	77	80
60	70	98	66	83

43	33	61	10	29
40	27	50	38	52
24	19	0	41	30
66	47	24	34	31
51	41	13	45	28

33	23	51	0	19
13	0	23	11	25
24	19	0	41	30
42	23	0	10	7
46	28	0	02	15

20	23	59	6	12
6	5	25	11	18
11	19	6	41	23
29	23	6	10	6
33	28	6	32	8

20	23	59	6	12
6	6	31	11	18
3	11	6	33	15
29	23	8	10	6
25	20	6	24	6

20	23	62	6	15
6	6	34	11	21
6	8	6	30	12
26	20	5	7	6
22	17	6	21	6

A → 4, B → 2, C → 1, D → 5, E → 3

$$101 + 84 + 87 + 80 + 98 = 450$$

Problem: 10

Solve the assignment Problem to min z the
result 3×4 matrix:

12	9	12	9
15	-	13	20
4	8	10	6

12	9	12	9
15	-	13	20
4	8	10	6
0	0	0	0

Now

3	6	3	0
2	-	0	7
0	4	6	2
0	0	0	0

$$9 + 13 + 4 + 0 = 26$$

6/8/2022

Problem : ①

Solve the following assignment and find minimum cost 5x6 matrix

12	3	6	-	5	8
4	11	-	5	-	3
8	2	10	9	7	5
-	7	8	6	12	10
5	8	9	4	6	-

Row

9	0	3	-	2	5
1	8	-	2	-	0
6	0	8	7	5	3
-	1	2	0	6	4
1	4	5	0	2	-
0	0	0	0	0	0

column

8	0	2	-	1	4
1	0	-	3	-	0
5	0	7	0	4	2
-	0	1	0	5	3
0	3	4	0	1	0
0	1	0	1	0	0

7	0	1	-	0	3
1	10	-	3	-	0
4	0	6	5	3	1
-	1	1	0	5	3
0	4	4	0	1	-
0	2	0	1	0	0

A → 5, B → 6, C → 2, D → 4;
~~E → 1, F → 3~~ E → 1, F → 3

$$5 + 2 + 0 + 6 + 5 + 3$$

$$= 21$$

Question Paper!

(1)

10	11	4	2	8
7	11	10	14	12
5	6	9	12	14
13	15	11	10	7
0	0	0	0	0

8	9	2	0	6
0	4	3	7	5
0	1	4	7	9
6	8	4	3	0
0	0	0	0	0

9	9	2	0	6
0	3	2	6	4
0	0	3	6	8
7	8	4	3	0
1	0	0	0	0

A → 4, B → 1, C → 2, D → 5, E → 3

2 + 7 + 6 + 7 + 0

= 22.

51	52	54	50
47	50	48	56
49	50	60	61
63	64	60	60

13	11	10	14
17	14	16	8
15	14	4	3
1	0	4	4

Row

3	1	0	4
9	6	8	0
12	11	1	0
1	0	4	4

OS + column

2	1	0	4
8	6	8	0
11	11	1	0
0	0	4	4

2	1	0	5
7	5	7	0
10	10	0	0
0	0	4	5

1	0	0	4
7	5	8	0
9	9	0	0
0	0	5	5

$$53 + 56 + 60 + 63 = 232$$

$\rightarrow A \rightarrow 2, B \rightarrow 4, C \rightarrow 3, D \rightarrow 1$

Graphically:

b) $\min Z = 2x_1 + 6x_2$

st: $4x_1 + 6x_2 \geq 4$

$x_1 + x_2 \leq 6$

$x_1 + 3x_2 \geq 5$

$x_1 + x_2 \geq 0$

$\Rightarrow 4x_1 + 6x_2 \geq 4$

x_1	0	1
x_2	0.7	0

$\Rightarrow x_1 + x_2 \leq 6$

x_1	0	6
x_2	6	0

$\Rightarrow x_1 + 3x_2 \geq 5$

x_1	0	5
x_2	1.6	0

$\min Z = 2x_1 + 6x_2$

$(5, 0)$

$\min Z = 2 \times 5 + 6 \times 0$
 $= 10$

$(6, 0)$

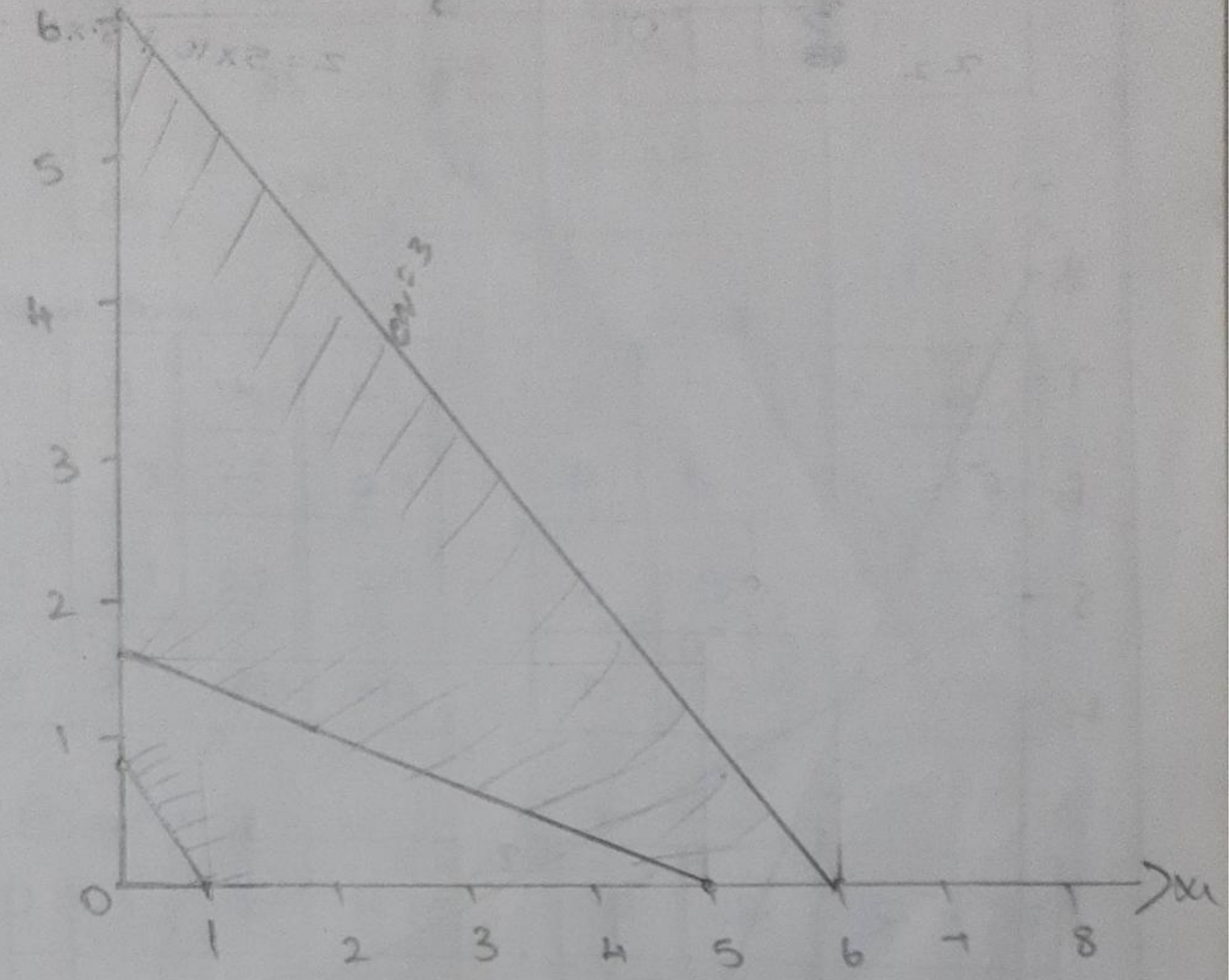
$\min Z = 2 \times 6 + 0$
 $= 12$

$(0, 6)$

$\min = 0 + 6 \times 6$

$= 36 \quad (0, 1.6)$

$$\begin{aligned} \text{Min } Z &= 5x_1 + 8x_2 = 5 \\ \text{s.t. } 2x_1 + x_2 &= 8 \\ x_1 + 2x_2 &= 10 \\ x_1, x_2 &\geq 0 \end{aligned}$$



$$\begin{aligned} \text{Min } Z &= 5x_1 + 8x_2 \\ \text{s.t. } 2x_1 + x_2 &\geq 8 \\ x_1 + 2x_2 &\geq 10 \\ x_1, x_2 &\geq 0 \end{aligned}$$

$$\Rightarrow 2x_1 + x_2 \geq 8$$
~~$$2x_1 + x_2 = 8$$~~

x_1	0	4
x_2	8	0

$$\begin{aligned} 2x_1 + x_2 &= 8 \\ x_1 = 0 \\ 2(0) + x_2 &= 8 \\ 0 + x_2 &= 8 \\ x_2 &= 8 \\ 2x_1 + 0 &= 8 \\ 2x_1 &= 8 \\ x_1 &= \frac{8}{2} \\ x_1 &= 4 \end{aligned}$$

$$\Rightarrow x_1 + 2x_2 \geq 10$$

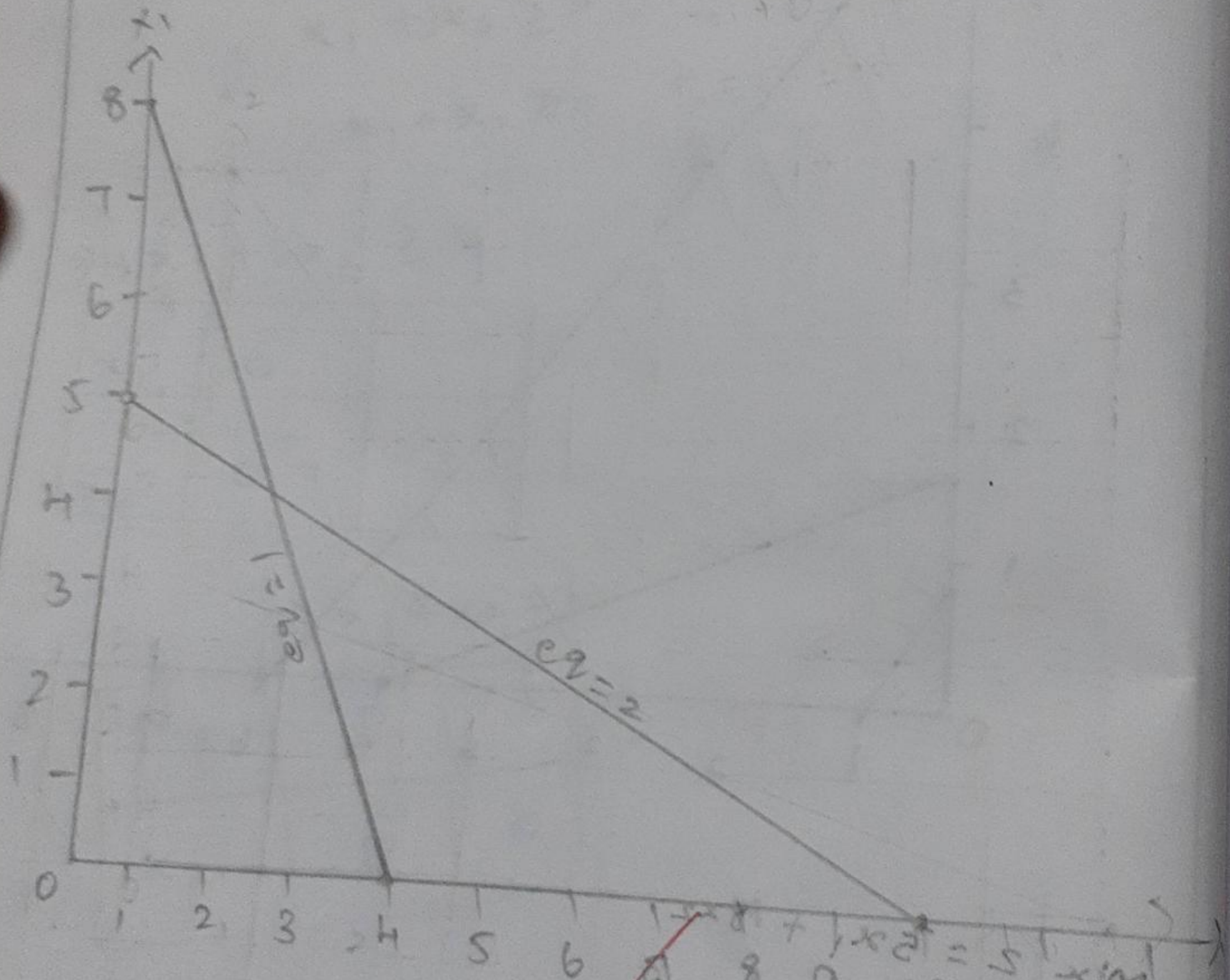
x_1	0	10
x_2	5	0

$$\min z = 5x_1 + 8x_2 = 64$$

$$z = 5x_1 + 8x_2 = 20$$

$$= 5x_1 + 8x_2 = 40$$

$$z = 5x_1 + 8x_2 = 50$$



$x_1 + 2x_2 = 10$
 $5x_1 + 8x_2 = 20$
 $5x_1 + 8x_2 = 40$
 $5x_1 + 8x_2 = 50$

$$\Rightarrow 5x_1 + 8x_2 \leq 64$$

H/w

① solve the assignment Problem Find max HXH.

42	35	28	21
30	25	20	15
30	25	20	15
24	20	16	12

Row Total

0	7	14	21
12	17	22	27
12	17	22	27
18	22	26	30

Row

0	7	14	21
0	5	10	15
0	5	10	15
0	4	8	12

Column

0	3	6	9
0	1	2	3
0	1	2	3
6	0	0	0

0	2	5	8
0	0	1	2
0	0	1	2
0	0	0	0

0	2	4	7
0	0	0	1
0	0	0	1
2	1	0	0

A → 1, B → 2, C → 3, D → 4
 $42 + 25 + 20 + 12 = 99$

TSM

(2)

∞	30	80	30
40	∞	140	30
40	50	∞	20
70	80	130	∞

the minimum cost

28 54

28 08

26 08

08 45

column

Row

∞	0	50	0
10	∞	110	0
20	30	∞	0
0	10	60	∞

∞	0	0	0
110	∞	60	0
20	30	∞	0
0	10	10	∞

∞	0	0	10
0	∞	50	0
10	20	∞	0
0	10	10	∞

∞	0	0	20
0	∞	40	0
10	10	∞	0
0	0	0	∞

A $\rightarrow$ C $\rightarrow$ D $\rightarrow$ B $\rightarrow$ A

$$80 + 20 + 80 + 40 = 220$$

40 0 28 54
15 0 15 0
PP = 51 + 02 + 25 + 54

1 0 0 0

4	6	10	5	6
7	4	-	5	4
-	6	9	6	2
9	3	7	2	3

4	6	10	5	6
7	4	-	5	4
-	6	9	6	2
9	3	7	2	3

Row

0	2	6	1	2
3	0	-	1	0
-	4	7	4	0
7	1	5	0	1
0	0	0	0	0

4	6	10	5	6
7	4	-	5	4
-	6	9	6	2
9	3	7	2	3
0	0	0	0	0

$$4 + 4 + 2 + 7 + 0 = 17$$

40	23	41	10
-	50	35	20
33	40	-	15
10	-	20	20

Row

30	13	31	0
-	30	15	0
8	35	-	0
0	-	10	20

Column

30	0	21	0
-	17	5	0
8	22	-	0
0	-	0	20

30	0	21	5
-	12	0	0
3	17	-	0
0	-	0	25

$$23 + 35 + 15 + 10 = 83$$

A → 2, B → 3, C → 4, D → 1

1)

Travelling Salesman Program:

	A	B	C	D
A	-	3	8	5
B	4	-	14	3
C	4	5	-	2
D	7	8	13	-

	2	3	8	5
4	∞	∞	14	3
4	6	∞	∞	2
7	8	13	∞	∞

Row

	2	3	8	5
1	∞	0	5	2
2	1	∞	11	0
3	2	3	∞	0
4	0	1	6	∞

Column

	2	3	8	5
1	∞	0	0	2
2	1	∞	6	0
3	2	3	∞	0
4	0	1	1	∞

	2	3	8	5
1	∞	0	0	3
2	0	∞	5	0
3	1	2	∞	0
4	0	1	1	∞

	2	3	8	5
1	∞	0	0	1
2	0	∞	4	0
3	1	1	∞	0
4	0	0	0	∞

$$A \rightarrow C \rightarrow B \rightarrow D \rightarrow A$$

$$8 + 5 + 3 + 7 = 23$$

$$A \rightarrow B \rightarrow D \rightarrow C \rightarrow A$$

$$3 + 3 + 13 + 14 = 23$$

$$14 + 0 + 2 + 3 + 8 = 27$$

$$14 + 0 + 2 + 3 + 8 = 27$$

3)

-	12	15	17	11
16	-	13	18	12
18	17	-	14	17
22	14	16	-	15
11	13	12	18	-

∞	1	12	15	17	11
16	∞	13	18	12	
18	17	∞	14	17	
22	14	16	∞	15	
11	13	12	18	∞	

Row

∞	1	4	6	0
4	∞	1	6	0
4	3	∞	0	3
8	0	2	∞	1
0	2	1	7	∞

column

∞	1	3	6	0
4	∞	0	6	0
4	3	∞	0	3
8	0	1	∞	1
0	2	1	7	∞

$A \rightarrow E \rightarrow A, B \rightarrow C \rightarrow D \rightarrow B, A \rightarrow E \rightarrow A$

∞	1	3	6	0
4	∞	0	6	0
4	3	∞	0	3
8	0	1	∞	1
0	2	1	7	8

$A \rightarrow E \rightarrow C \rightarrow D \rightarrow B \rightarrow A$

$$11 + 12 + 14 + 14 + 16 = 67.$$

4)

A	-	1	6	8	H
B	7	-	8	5	6
C	6	8	-	9	H
D	8	5	9	-	8
E	4	6	7	8	-

Row

∞	0	5	7	3
2	∞	3	0	1
10	2	∞	3	1
3	0	4	∞	3
0	2	3	4	∞

∞	0	1	6	1
2	∞	∞	0	∞
∞	2	∞	3	0
3	∞	0	∞	1
0	2	∞	4	∞

Column

∞	0	2	7	2
2	∞	0	0	∞
0	2	∞	3	∞
3	∞	1	8	∞
∞	2	0	4	8

A → B → D → C → E → A

$$1 + 5 + 7 + 9 + H = 26$$

5)

-	12	7	6	5	5
9	-	13	5	13	10
6	13	-	7	10	8
4	9	10	-	6	9
5	13	7	6	-	4
5	11	9	6	5	-

row

∞	67	2	1	0	0
4	∞	8	0	8	5
0	7	∞	1	4	2
0	5	6	8	2	5
1	9	3	2	∞	0
0	6	4	1	0	∞

column

∞	2	0	1	∞	∞
4	∞	6	0	8	5
0	2	∞	1	4	2
∞	0	3	∞	2	5
1	4	1	2	∞	0
0	1	2	1	∞	∞

$A \rightarrow C \rightarrow A, B \rightarrow D \rightarrow B, E \rightarrow F \rightarrow E$

∞	2	0	1	0	0
∞	∞	6	0	8	5
0	2	∞	1	4	2
0	0	4	∞	2	5
1	4	1	2	∞	0
0	1	2	1	0	∞

$A \rightarrow F \rightarrow E \rightarrow C \rightarrow D \rightarrow B \rightarrow A$

$$5 + 5 + 7 + 7 + 9 + 9$$

$$= 42.$$

$E \rightarrow C \rightarrow B \rightarrow D \rightarrow A \rightarrow F \rightarrow E$

6)

-	7	5	3	5
7	-	8	4	3
5	8	-	6	2
3	4	6	-	2
5	3	2	2	-

2	2	2	7	5
01	21	2	21	-
8	01	7	-	21
P	2	-	01	P
H	-	2	7	21
-	2	2	P	11

columns

8	4	2	0	2
4	8	5	1	0
3	6	8	4	0
1	2	4	8	0
3	1	0	0	8

8	3	2	0	2
3	8	5	1	2
2	5	8	4	0
0	1	4	8	0
2	0	0	0	8

$A \rightarrow D \rightarrow B \rightarrow E \rightarrow C \rightarrow A$

$$3 + 4 + 3 + 2 + 5 = 17$$

ACAGAG
PTATTTT

Problem ①

A Person requires atleast 10, 12, 12 units of chemical A, B, C liquid Product contain 5, 2 and 1 units of A, B, C respectively Per Tin. A dry Product contain 1, 2, 4 units of A, B, C Per cotton box the liquid Product sells for ₹ 3 Per dry Product sell for ₹ 2 Per cotton box?

x_1 Be the no of ^{liquid} units A, B & C

x_2 Be the no of dry units A, B & C

$$\text{Min } Z = 3x_1 + 2x_2$$

St ^y	A	B	C
liquid:	5	2	1
dry:	1	2	4
	10	12	12

₹ 3 Per Jar

₹ 2 / cotton

$$5x_1 + x_2 \geq 10$$

$$2x_1 + 2x_2 \geq 12$$

$$x_1 + 4x_2 \geq 12$$

$$x_1 + x_2 \geq 0$$

equ ①: $5x_1 + x_2 \geq 10$

x_1	0	2
x_2	10	0

Min $Z = 3x_1 + 2x_2$

$3 \times 3 + 2 \times 3 = 15$

$3 \times 2 + 2 \times 4 = 14$

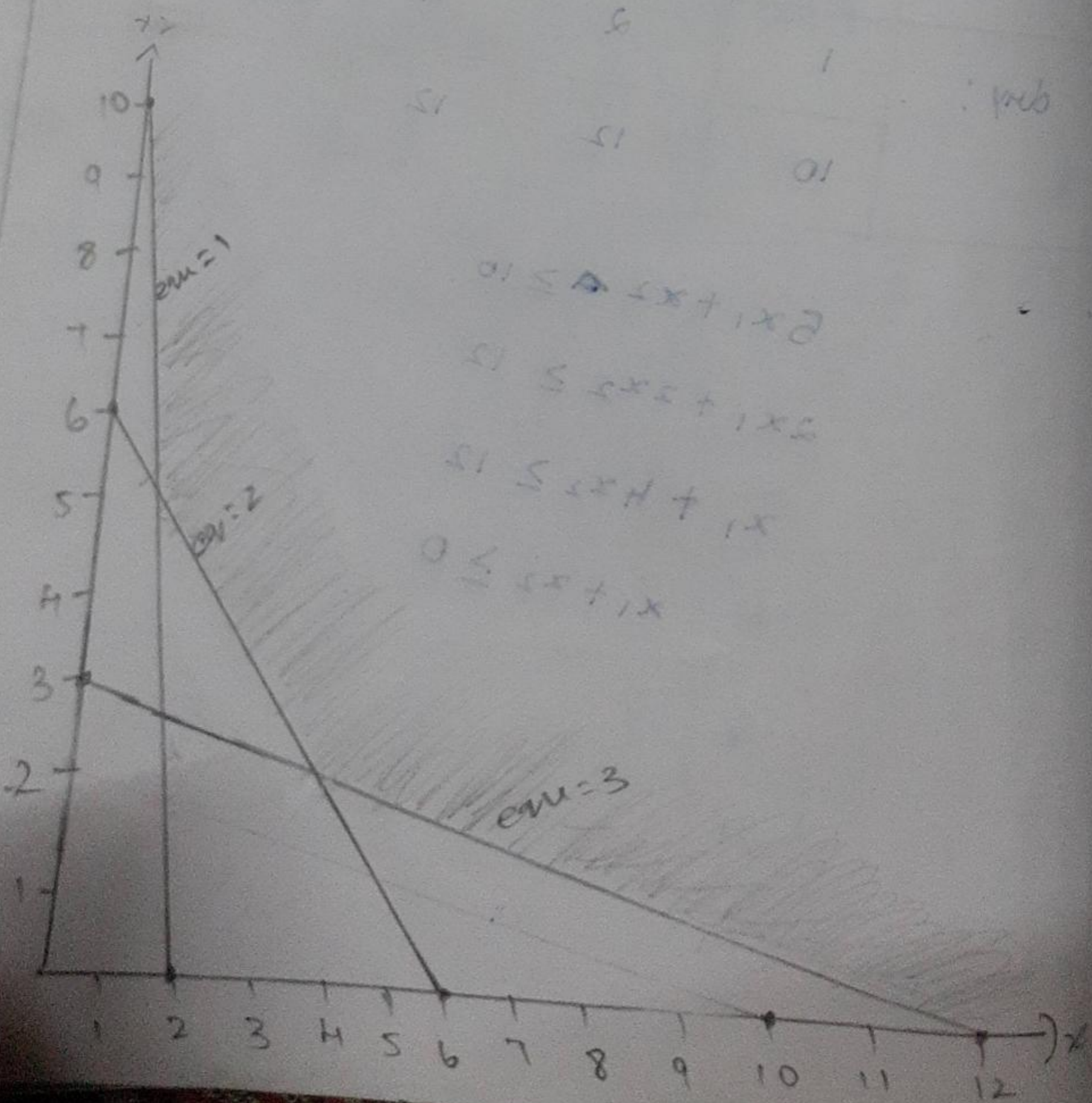
$x_1 = 2, x_2 = 4, Z = 14$

equ ②: $2x_1 + 2x_2 \geq 12$

x_1	0	6
x_2	6	0

equ ③: $x_1 + 4x_2 \geq 12$

x_1	0	12
x_2	3	0



Home work:

graphically

$$\min Z = 6x_1 + 3x_2$$

$$s.t.: 4x_1 + 3x_2 \leq 6$$

$$x_1 + 4x_2 \geq 4$$

$$x_1 + x_2 \geq 0$$

2) Solve graphically:

$$\min Z = x_1 + x_2$$

$$s.t.: x_1 + 3x_2 \geq 4$$

$$x_1 + 6x_2 \geq 8$$

$$x_1 + x_2 \geq 0$$

$$3) \min Z = 2x_1 + 4x_2$$

$$s.t.: 2x_1 + 2x_2 \geq 4$$

$$x_1 + 3x_2 \geq 3$$

$$2x_1 + 4x_2 \geq 2$$

$$x_1 + x_2 \geq 0$$

4) solve

10	21	30	28
11	31	42	23
41	18	16	26

the assignment and max the

Profit:

5) consider the Problem of assigning 5 operators

5 machine. 5 mil

7	7	-	4
9	6	4	5
11	5	7	-
9	4	8	9
8	7	9	11

6) TSM:

-	12	24	25	15
6	-	16	18	7
10	11	-	18	12
14	17	22	-	16
12	13	23	25	-

$$4x_1 + 3x_2 = 6$$

$$0 + 3x_2 = 6$$

$$x_2 = \frac{6}{3} = 2 \quad (0, 2)$$

$$4x_1 + 3x_2 = 6$$

$$4x_1 + 0 = 6$$

$$x_1 = \frac{6}{4} = 1.5 \quad (1.5, 0)$$

$$x_1 + Hx_2 = H$$

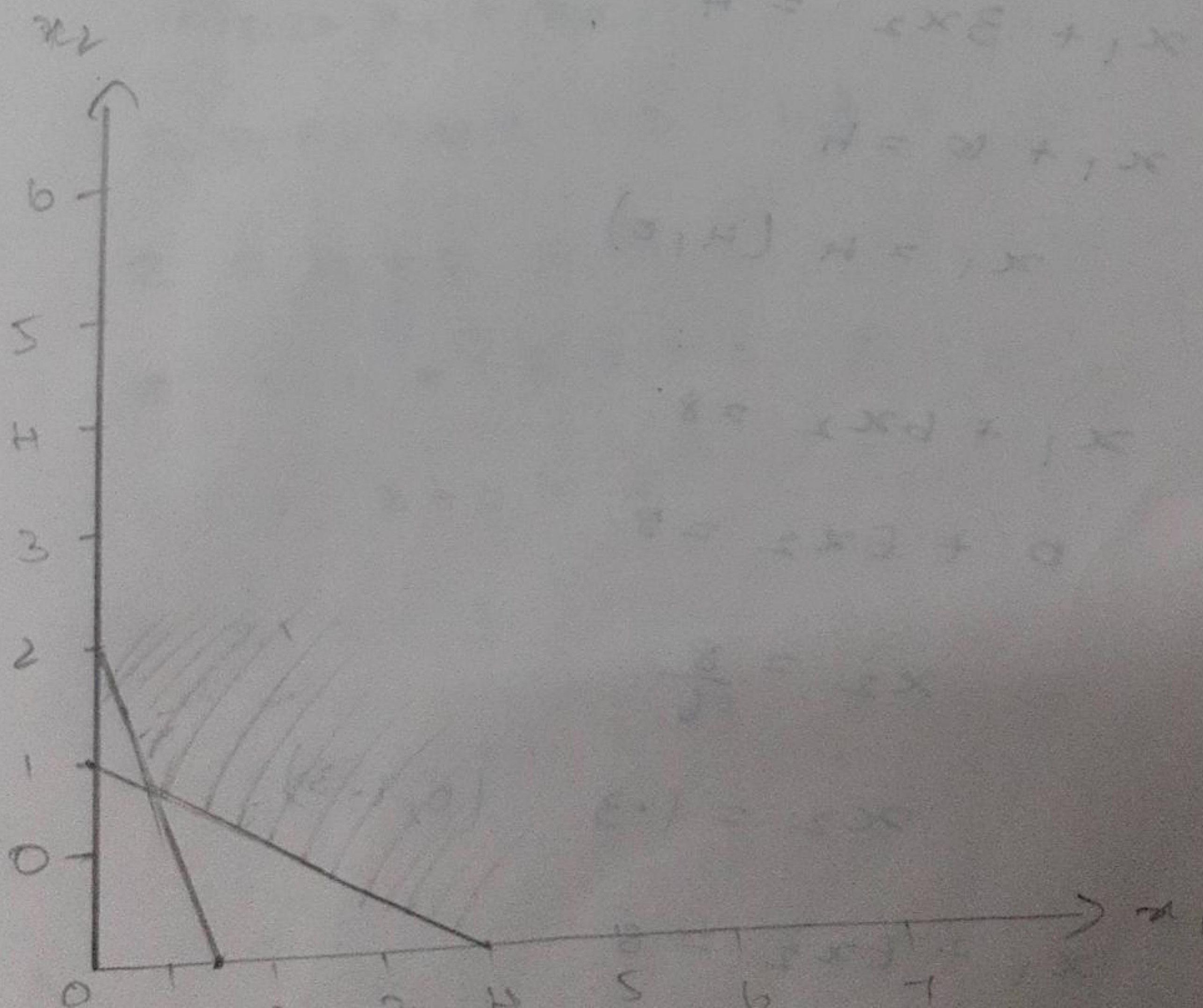
$$0 + Hx_2 = H$$

$$x_2 = \frac{H}{H} = 1 \quad (0, 1)$$

$$x_1 + Hx_2 = H$$

$$x_1 + 0 = H$$

$$x_1 = H \quad (H, 0)$$



$$111 = 0x_1 + 3x_2$$

$$2 = 0x_0 + 3x_2$$

$$2 = 0 + 6 = 6$$

$$2 = 6 \times 15 + 3 \times 0 = 9$$

$$2 = 6 \times 0 + 3 \times 1 = 3$$

$$2 = 6 \times 4 + 3 \times 0 = 24$$

$$x_1 = 4, x_2 = 0, 2 = 24$$

$$2) \quad x_1 + 3x_2 = 4$$

$$0 + 3x_2 = \frac{4}{3}$$

$$x_2 = 1.3 \quad (0, 1.3)$$

$$x_1 + 3x_2 = 4$$

$$x_1 + 0 = 4$$

$$x_1 = 4 \quad (4, 0)$$

$$x_1 + 6x_2 = 8$$

$$0 + 6x_2 = 8$$

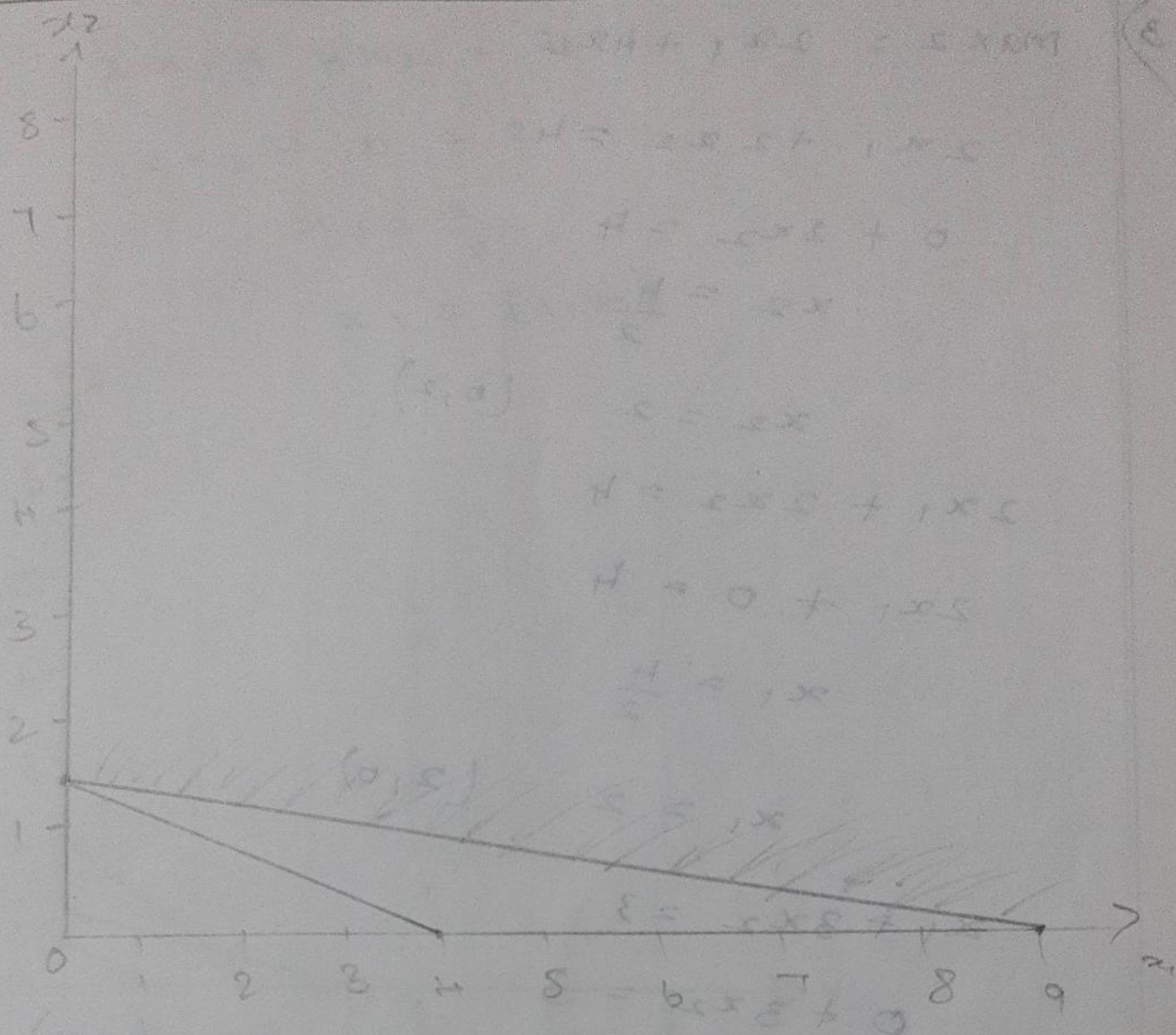
$$x_2 = \frac{8}{6}$$

$$x_2 = 1.3 \quad (0, 1.3)$$

$$x_1 + 6x_2 = 8$$

$$x_1 + 0 = 8$$

$$x_1 = 8 \quad (8, 0)$$



$$\min z = x_1 + x_2$$

$$z = x_1 \cdot 0 + 1.3 = 1.3$$

$$z = 4 \cdot 0 = 4$$

$$z = 0 \cdot 1.3 = 1.3$$

$$z = 8 + 0 = 8$$

$$\min z = 1.3$$

$$x_1 = 0, x_2 = 1.3$$

3)

$$\max z = 2x_1 + 4x_2$$

$$2x_1 + 2x_2 = 4$$

$$0 + 2x_2 = 4$$

$$x_2 = \frac{4}{2}$$

$$x_2 = 2$$

(0, 2)

$$2x_1 + 2x_2 = 4$$

$$2x_1 + 0 = 4$$

$$x_1 = \frac{4}{2}$$

$$x_1 = 2$$

(2, 0)

$$x_1 + 3x_2 = 3$$

$$0 + 3x_2 = 3$$

$$x_2 = \frac{3}{3}$$

(0, 1)

$$x_1 + 3x_2 = 3$$

$$x_1 + 0 = 3$$

$$x_1 = 3$$

(3, 0)

$$2x_1 + 4x_2 = 2$$

$$0 + 4x_2 = 2$$

$$x_2 = \frac{2}{4}$$

$$x_2 = 0.5$$

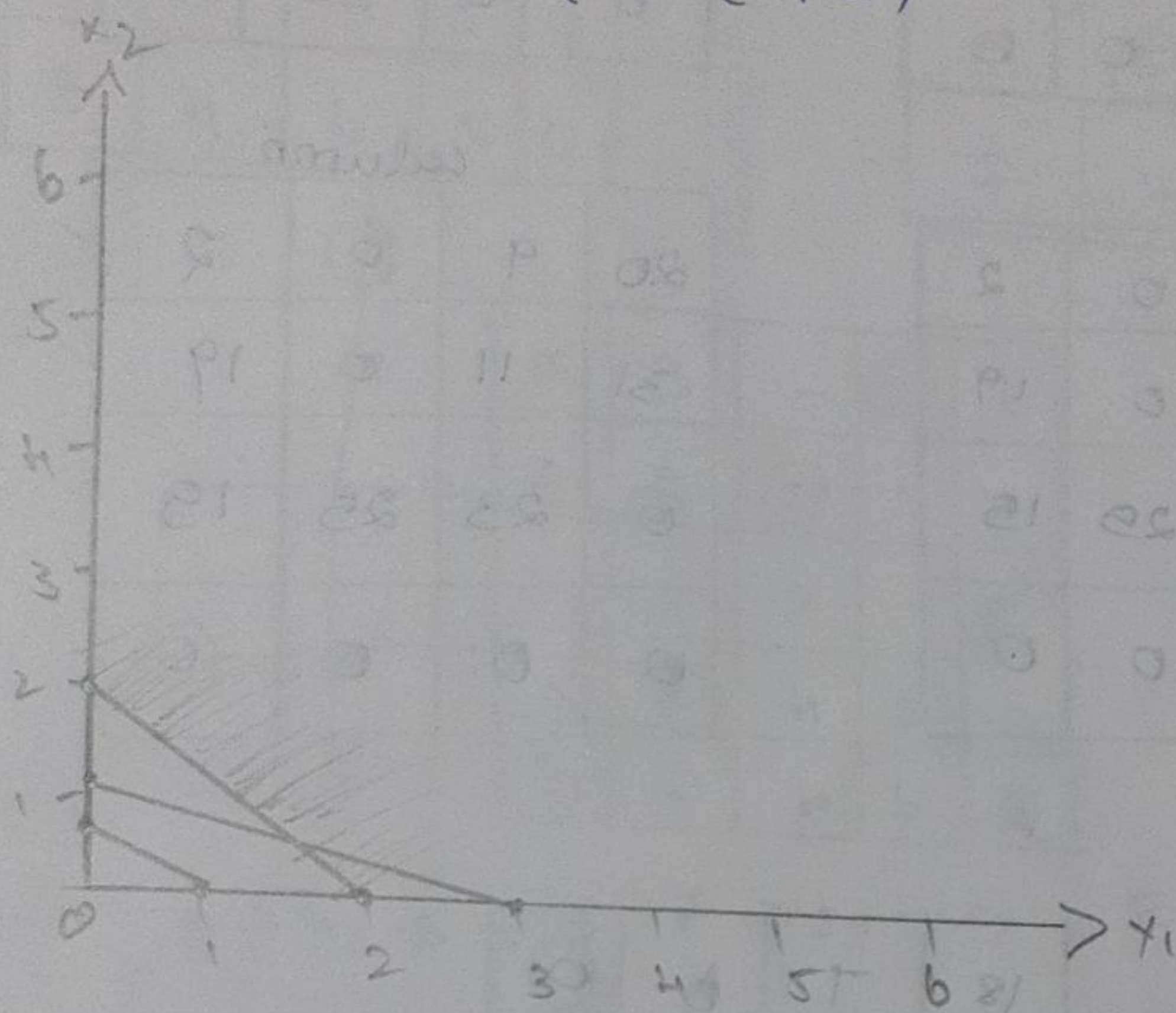
(0, 0.5)

$$2x_1 + 4x_2 = 2$$

$$2x_1 + 0 = 2$$

$$x_1 = \frac{2}{2}$$

$$x_1 = (1, 0)$$



$$\text{Max } Z = 2x_1 + 4x_2$$

$$Z = 2 \times 0 + 4 \times 2 = 8$$

$$Z = 2 \times 2 + 4 \times 0 = 4$$

$$Z = 2 \times 0 + 4 \times 1 = 4$$

$$Z = 2 \times 3 + 4 \times 0 = 6$$

$$Z = 2 \times 0 + 4 \times 0.5 = 2$$

$$Z = 2 \times 1 + 4 \times 0 = 2$$

$$\text{Max } Z = 8$$

$$x_1 = 0, x_2 = 2$$

A)

10	21	30	28
11	31	42	23
41	18	16	26
0	0	0	0

32	21	12	14
31	11	0	19
1	24	26	16
0	0	0	0

row

20	9	0	2
31	11	0	19
0	23	25	15
0	0	0	0

column

20	9	0	2
31	11	0	19
0	23	25	15
0	0	0	0

18	7	0	0
29	9	0	17
0	23	27	15
0	0	2	0

 $A \rightarrow B, B \rightarrow C, C \rightarrow A, D \rightarrow B$

$$28 + 42 + 41 + 0 = 111.$$

Row				
3	3	0	0	4
5	2	0	1	2
6	0	2	0	0
5	0	4	5	0
5	4	6	8	0

column				✓
①	3	0	8	1
2	2	⑤	1	2
3	0	2	-	0
2	⑤	4	5	0
2	4	6	8	0

0	3	-	⑤	6
2	2	⑤	1	4
1	0	2	-	⑤
2	⑤	4	5	2
⑤	2	4	6	0

$A \rightarrow 4, B \rightarrow 3, C \rightarrow 5, D \rightarrow 2, E \rightarrow 1.$

$$4 + 4 + 5 + 4 + 8 = 25.$$

Row				
8	0	12	13	3
0	0	10	12	1
0	1	0	8	2
0	3	8	0	2
0	1	11	13	0

column				
0	⑤	4	5	2
0	0	②	4	0
0	1	0	⑤	1
0	3	0	0	①
⑤	1	3	5	0

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow A$

$$12 + 16 + 18 + 16 + 12 = 74$$

Transportation Problem!

$$\rightarrow \min Z = \sum_{i=1}^m \sum_{j=1}^n C_{ij} x_{ij}$$

$M =$ Source.

$N =$ Destination.

St:

$$\sum_{j=1}^n x_{ij} = a_i \quad i = 1 \text{ to } m$$

$$\sum_{i=1}^m x_{ij} = b_j \quad j = 1 \text{ to } n$$

$$i = 1 \text{ to } m \text{ \& } j = 1 \text{ to } n$$

$$x_{ij} \geq 0$$

Problem ! ①

$i \backslash j$	1	2	3	
1	2	7	4	5
2	3	3	1	8
3	5	4	7	7
4	1	6	2	14
				<u>34</u>

Supply: 7 9 18 34

NWCR: North west corner

$i \backslash j$	1	2	3	
1	2	7	4	5
2	3	3	1	8
3	5	4	7	7
4	1	6	2	14
				<u>34</u>

7 9 18

3	4	3	1	86
5		4	7	7
1		6	2	14
		0	18	24

3	4	3	80
4		7	7
6		2	14

4	4	7	74
		2	14

7	4	40
2		14

2	40
	0

$$2 \times 5 + 3 \times 2 + 3 \times 6 + 4 \times 3 + 7 \times 4 + 2 \times 1$$

$$= 10 + 6 + 18 + 12 + 28 + 2$$

$$= 102$$

~~Definition~~ Definition:

feasible solution to a transportation problem
 a set of non negative allocation x_{ij}
 that satisfy row and column
 condition.

Basic Feasible Solution:

Its set to be not more than $m + 1$ not negative allocation.

where m is number of rows
 n is number of columns

non-degenerate Basic Feasible Solution:

A Basic Feasible Solution to a $\frac{m}{n}$ constraint problem is set to be non de , if

→ The Total number of non negative allocation is $m + n - 1$ and

→ $m + n - 1$ allocations on independent positions

Degenerate Basic Feasible Solution:

A basic feasible solution to a transportation problem

no of non negative allocation is less than $m + n - 1$ is called

~~degenerate~~, DBFS.

Optimal Solution:

A feasible solution that ~~min~~ ^{min} the a transportation cost or max the cost is called Solution.

Problem: (2)

13	10	15	200
10	12	9	400
25	30	18	400
			<u>1000</u>

Supply: 200

13	10	15	200
10	12	9	400
25	30	18	400
			<u>800</u>

12	300	9	100
30		18	400
			<u>500</u>

9	100	400
18	400	800
		<u>1200</u>

18	400	1600
		<u>1600</u>

Work

$$13 \times 200 + 12 \times 300 + 9 \times 100 + 18 \times 400$$

$$= 6000 + 3600 + 900 + 7200$$

$$= 17700$$

③

H	6	8	2	10
10	13	15	7	20
11	16	18	8	50

Demand: 15 15 25 25 80

H	6	8	2	10
10	13	15	7	20
11	16	18	8	50

10	13	15	7	15
11	16	18	8	50

13	15	7	150
16	18	8	50

18	8	25
----	---	----

8	25
---	----

$$\begin{aligned}
 & H \times 10 + 10 \times 5 + 13 \times 15 + 18 \times 25 + 8 \times 25 \\
 & = 10H + 50 + 195 + 450 + 200 \\
 & = 935
 \end{aligned}$$

④

H	8	2	1	7
6	7	5	1	9
8	9	10	2	2

Demand: 9 1 H H 18

H	8	2	1	7
6	7	5	1	9
8	9	10	2	2

6	2	7	5	1	9
8	9	10	2	2	

20 1 4 4

7	1	5	1
9		10	2

5	4	1
10		2

1	2
2	

2	2
---	---

$$\begin{aligned}
 & 4 \times 7 + 6 \times 2 + 7 \times 1 + 5 \times 4 + 2 \times 2 \\
 & = 28 + 12 + 7 + 20 + 4 \\
 & = 73
 \end{aligned}$$

⑤

21	32	52	12	7000
72	32	42	62	9000
42	10	72	22	18000

Demand: 5000

21	32	52	12
72	32	42	62
42	10	72	22

Demand: 5000

32	52	12
32	42	62
10	72	22

8000 7000 14000
6000

8000
7000
9000
18000
34000

32	42	62	4000
10	72	22	15000
6000	7000	14000	

42	62	3000
72	22	18000
7000	14000	

72	22	18000	14000
7000	14000		

22	14000	14000
14000		

$$\begin{aligned}
 & 3 \times 2000 + 32 \times 6000 + 42 \times 3000 + 72 \times 4000 + 22 \times 14000 \\
 &= 108000 + 64000 + 192000 + 126000 + 308000 \\
 &= 1083000
 \end{aligned}$$

6)

LCM $\Rightarrow$ Least cost method

3	4	2	8
5	6	8	2
10	12	7	10

Demand: 3 7 10 20

3	4	2	8
5	6	8	2
10	12	7	10

5	2	6	8
10	12	7	10

10	12	7	10
----	----	---	----

10	12	8	7
----	----	---	---

12	7	10
----	---	----

$$2 \times 8 + 5 \times 2 + 7 \times 2 + 10 \times 1 + 12 \times 7$$

$$= 16 + 10 + 14 + 10 + 84$$

$$= 134$$

~~regulation method~~

②

4	6	8	12	10
10	13	15	7	20
11	16	18	8	

D: 15 15 25 25 80

4	6	8	2	10
10	13	15	7	20
11	16	18	8	50

15 15 25 ~~25~~
15

10	13	15	7
11	16	18	8

15 45 25 15
8 2 2 0

10	5	13	15
11		16	18

15 15 25 50
10

10	10		
11	16	18	

10 15 25
0

16	15	18
---------------	----	----

15 25 40
0

18	25
----	----

25 250
0

$$2 \times 10 + 7 \times 15 + 10 \times 5 + 11 \times 10 + 16 \times 15 + 18 \times 25$$

$$= 20 + 50 +$$

$$= 975$$

Home work

③

6	1	2	10
4	3	0	8
5	7	9	2
4	2	1	5

D: 10

12

3 25

④

10	12	4	3	8
8	9	7	6	2
12	4	8	2	12

D: 7

9

4

2 22

⑤

10	25	27	10
34	33	12	17
60	52	14	

D: 15

7

8

30

③

6	1	10	2	10 ⁰	4	3	0	3	8	5	0
4	3		0	8	5	7	9	2			
5	7		9	2	4	2	1	5			
4	2		1	5	10	2	30				

4		3	2	3
5		7	2	
4		2	5	
10		70		

4	3	30
5		2
4		5
10		7

5	2	20
4	5	30
4	2	8

$$5^2 = 20$$

$$1 \times 10 + 0 \times 3 + 3 \times 2 + 4 \times 3 + 4 \times 5 + 5 \times 2$$

$$= 10 + 0 + 6 + 12 + 20 + 20$$

$$= 56$$

④

10	12	4	3	8
8	9	7	6	2
12	4	8	2	12
7	9	4	20	

10	12	4	4	8
8	9	7		2
12	4	8		10
7	9	4		

10	12	H
8	9	2
12	H	10
7	90	

10	H
8 2	70
12	1
75	02

10 H	H 0
12	1
31	2

12	1	10
H	10	

$$= 2 \times 2 + H \times H + H \times 9 + 8 \times 2 + H \times 10 + H \times 10 + 12 \times 1$$

$$= H + 16 + 36 + 16 + H0 + 12$$

$$= 12H.$$

⑤

10	10	25	27	100
3H	33	12	17	
60	53	H1	3	
155	7	8		

3H	33	12	8	179
60	53	H1	3	
5	7	80		

3H	33	17	92
60	53	3	
5	70		

3H	2	20
60	3	
83		

60	3	80
80		

$$= 10 \times 10 + 12 \times 8 + 33 \times 7 + 3H \times 2 + 60 \times 3$$

$$= 100 + 96 + 231 + 68 + 180$$

$$= 675$$

Row minima method:

7	9	3	2	16
4	4	3	5	14
6	4	5	9	20
D:	11	9	25	8
				<u>50</u>

7	9	3	16
4	4	3	14
6	4	5	20
11	9	25	8

7	9	3	19
4	4	3	11
6	4	5	15
11	9	25	45

4	4	3	14
6	4	5	20

→

6	4	20		
11	9	25	8	50

6	20			
11	9	25	8	50

$$3 \times 8 + 2 \times 8 + 3 \times 4 + 4 \times 9 + 6 \times 11$$

$$= 24 + 16 + 12 + 36 + 36$$

$$= 184$$

②

4	6	8	2	6	
8	7	6	4	16	
3	2	6	7	4	
1	0	2	4	36	
D:	12	8	25	5	<u>50</u>

4	6	8	2	5
8	7	6	4	
3	2	6	7	
1	0	2	4	
12	8	25	30	

4	1	6	8	10
8		7	6	10
3		2	6	4
1		0	2	30
12	11	2	25	50

8	7	6	60
3	2	6	4
1	0	2	30
12	8	25	5

3	2	H	6	10
1	0	2	2	30
11	8	15		

1	0	H	2	26
11	0	15		

1	11	2	2	26
11	15			

15	2	15		
----	---	----	--	--

$$\begin{aligned}
 & H \times 1 + 2 \times 5 + 6 \times 10 + 2 \times H + 0 \times H + 1 \times 11 + 2 \times 15 \\
 & = H + 10 + 60 + 8 + H + 11 + 30 \\
 & = 123
 \end{aligned}$$

10	12	18	8	106
6	7	3	2	150
1	18	8	7	50
				<u>300</u>

D: 150

100

25

25

H	8	7	9	8
6	8	3	2	2
10	2	3	6	50
				<u>60</u>

D: 10

20

25

5

Column Minima Method !

①

D:

10	21	8	10
	15	20	20
11		11	30
3	8		60
15	30	15	

10	21	8	10
	15	20	20
11		11	30
3	8		15
15	30	15	

21	8	10
15	20	20
8	11	15
30	15	

21	8	10
15	20	20
15		5

15	20	5

20	5	50

$$m + n - 1 = 5$$

$$3 \times 15 + 8 \times 15 + 8 \times 10 + 15 \times 15 + 20 \times 5$$

$$= 45 + 120 + 80 + 225 + 100$$

$$= 570 //$$

P	T	8	H
5	8	8	10
2	8	5	10

Row Minima method

H/w

3

10	12	18	8	25	100	75
6	7	3	2		150	
1	18	8	7		50	
150	100	25	25	0		

10	75	12	18	75 0
6	7	3		150
1	18	8		50
150	100	25		75

6	7	3	25	150	125
1	18	8		50	
75	100	25	0		

6	75	7	125
1		18	50
75	0	100	

7	75	150 0
18		50
100	25	

1	8	50 25
		25
		0

B.A.M. Vogel's Approximation Method

4	8	3	0	16
16	24	16	0	52
8	16	24	0	48
12	12	24	20	235

4	8	3	0	16 (4)
16	24	16	0	52 (8)
8	16	24	0	48 (8)
12	12	24	20	0
(4)	(8)	(8)	0	

4	8	3	0	16 (4)
16	24	16	0	52 (8)
8	16	24	0	48 (8)

16	24	16	0	52 (8)
8	16	24	0	48 (8)
0	20	24	20	0
(8)	(8)	(8)		

16 (4)
24 (8)
24 (8)

24	16	0	16 (4)
16	24	0	52 (8)
20	14	0	
(8)	(8)		

24	16	0	16 (4)
16	24	0	52 (8)
20	14	0	
(8)	(8)		

16 5 50

$$m = 3$$

$$n = 4$$

$$3 + 4 - 1 = 6$$

$$\begin{aligned} \min Z &= 0 \times 80 + 8 \times 16 + 8 \times 72 + 24 \times 20 + 16 \times 4 + 3 \times 12 \\ &= 0 + 128 + 576 + 480 + 64 + 36 \\ &= 2424 \end{aligned}$$

②

10	12	15	22	11	10
25	16	15	12	10	25
14	17	19	20	25	5
					<u>40</u>
D: 10	5	3	2	20	

10	12	15	22	11	10 (1)
25	16	15	12	10	23 (2)
14	17	19	20	25	5 (3)
10	5	3	20	20	
(14)	(14)	(10)	(8)	(1)	

10	12	15	11	10 (1)
25	16	15	10	23 (5)
14	17	19	25	5 (3)
10	5	3	20	
(14)	(14)	(10)	(1)	

10	12	15	10 (2)
25	16	15	3 (1)
14	17	19	5 (3)
10	5	3	
(14)	(14)	(10)	

16	15	3 (1)
17	19	3 (2)
50	3	
(1)	(14)	

15	3	20	(1)
19		0	(2)

20
(1)

$$m+n=1$$

$$3+5-\frac{1}{2}=7$$

$$= 12 \times 2 + 10 \times 2 + 10 \times 10 + 14 \times 4 + 15 \times 4 + 15 \times 13 + 17 \times 13$$

$$= 24 + 200 + 100 + 0 + 56 + 195 + 221$$

$$= 454$$

②

10	15	12	12	200
8	10	11	19	150
11	12	13	10	120
100	120	30	200	370

10	15	12	12	200 (2)
8	10	11	19	150 (1)
11	12	13	10	120 (1)
0	0	0	0	00 (0)
100	120	30	200	370
(8)	(10)	(11)	(19)	

15	18	12	200 (1)
10	11	19	100 (1)
12	15	10	120 (2)
0	0	0	00 (0)
100	200	200	
(10)	(11)	(19)	

15	12	12	200
10	11	19	100
12	13	10	120

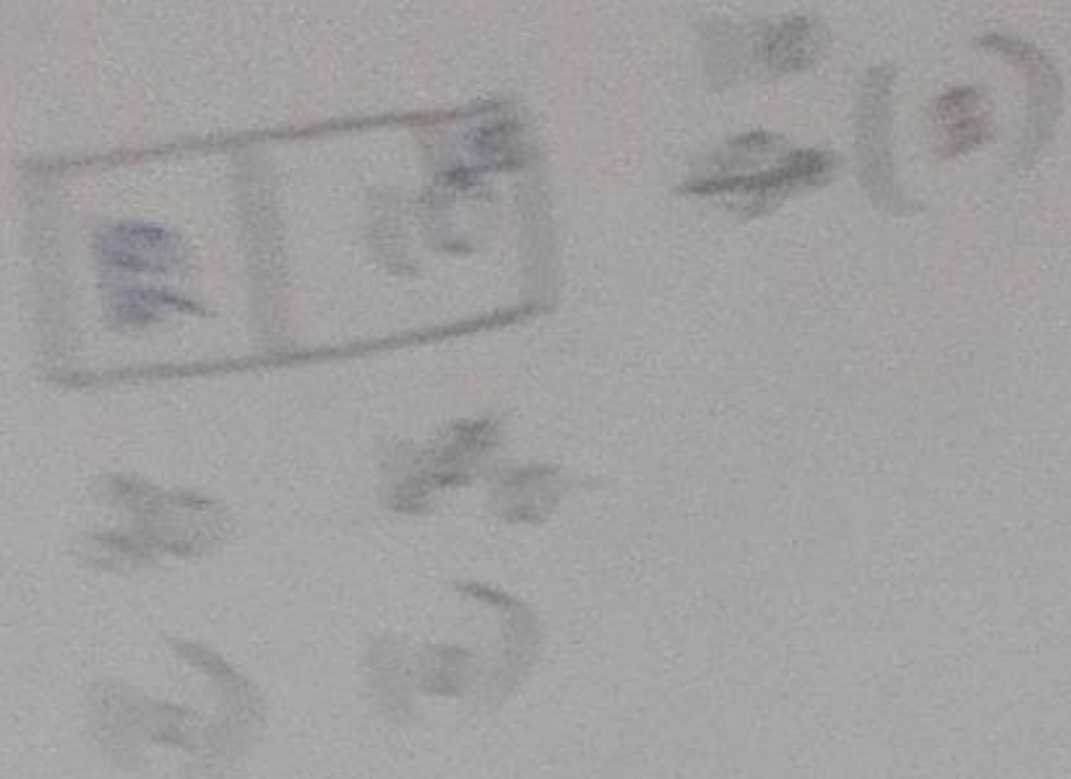
15	12	200 (3)
10	19	100 (1)
12	10	120 (2)
0	0	00 (0)

18
15

15	12	200 (3)
10	19	100 (1)
12	10	120 (2)
20	200	200
(2)	(2)	

15	12	200 (3)
10	10	120 (2)
20	200	200
(3)	(2)	

(c)



[Faint, illegible text]

[Faint, illegible text]

[Faint, illegible text]

30	25	40	20	100
29	26	35	40	250
31	33	37	30	150
90	160	200	50	500

30	25	40	20	100
29	26	35	40	250
31	33	37	30	150
90	160	200	50	
(1)	(1)	(2)	(10)	

30	25	40	0
29	26	35	250(3)
31	33	37	150(2)
90	160	200	
(1)	(1)	(2)	

29	26	35	140(3)
31	33	37	150(2)
90	140	200	
(2)	(7)	(2)	

29	90	35	50(6)
31		37	150(6)
200	200		
(2)		(2)	

35	50
37	150
200	
(2)	

35	50
50	

$$M+n-1$$

$$3+4-1=6$$

$$= 25 \times 50 + 26 \times 110 + 29 \times 90 + 37 \times 150 + 35 \times 50$$

=

VAM

5) Find the optimal transportation cost.

2	7	4	5 (2)
3	3	7	8 (0)
5	4	1	7 (3)
1	6	2	14 (1)

Supply : 7 9 18
(1) (1) (1)

2	7	4	5 (2)
3	3	7	8 (0)
5	4	1	7 (3)
1	6	2	14 (1)

2	5	7	4	5 (2)
3	3	7		80 (0)
1	6	2		14 (1)

7 9 11

(1) (3) (2)

2	7	4	5 (2)
1	6	2	11 (1)

5	7	0 (5)
1	6	3 (5)

2	6	5 (5)
---	---	-------

$$\boxed{64} \quad Y_0$$

X_0

$$= 1 \times 7 + 3 \times 8 + 2 \times 11 + 2 \times 5 + 1 \times 1 + 6 \times 1$$

$$= 7 + 24 + 22 + 10 + 1 + 6$$

$$= 71$$

UV Method

2	$\boxed{4}$	7	$\boxed{1}$	H	$V_1 = 0$
3		3	$\boxed{8}$	7	$V_2 = 4$
5		H		1	$V_3 = -2$
1	$\boxed{3}$	6		2	$UV = -1$

$$V_1 = 2 \quad V_2 = 7 \quad V_3 = 3$$

$$C_{11} = U_1 + V_1$$

$$2 = 0 + V_1$$

$$\boxed{V_1 = 2}$$

$$C_{12} = U_1 + V_2$$

$$7 = 0 + V_2$$

$$\boxed{V_2 = 7}$$

$$C_{33} = V_2 + V_2$$

$$3 = V_2 + 4$$

$$V_2 = 3 - 4$$

$$\boxed{V_2 = -1}$$

$$C_{33} = V_3 + V_3$$

$$\boxed{1 = V_3 + V_3} \rightarrow \text{①}$$

$$1 = V_3 + 3$$

$$V_3 = 1 - 3$$

$$= -2$$

$$C_{41} = V_1 + V_1$$

$$1 = V_1 + 2$$

$$V_1 = 1 - 2 = -1$$

$$\boxed{V_1 = -1}$$

$$C_{43} = V_1 + V_3$$

$$2 = -1 + V_3$$

$$V_3 = 2 + 1$$

$$\boxed{V_3 = 3}$$

$$\boxed{d_{13} = C_{13} - V_1 - V_3}$$

$$d_{13} = 4 - V_1 - V_3$$

$$= 4 - 0 - 3$$

$$\boxed{d_{13} = 1}$$

$$d_{21} = L_{21} - v_2 - v_1$$

$$= 3 - (-4) - 2$$

$$= 3 + 4 - 2$$

$$d_{21} = 8$$

$$d_{23} = L_{23} - v_2 - v_3$$

$$= 7 - (-4) - 3$$

$$d_{23} = 8$$

$$d_{31} = L_{31} - v_3 - v_1$$

$$= 5 - (-2) - 2$$

$$d_{31} = 5$$

$$d_{32} = L_{32} - v_3 - v_3$$

$$= 4 - (-2) - 2$$

$$d_{32} = -1$$

$$d_{42} = L_{42} - v_4 - v_2$$

$$= 6 - (-1) - 7$$

$$d_{42} = 0$$